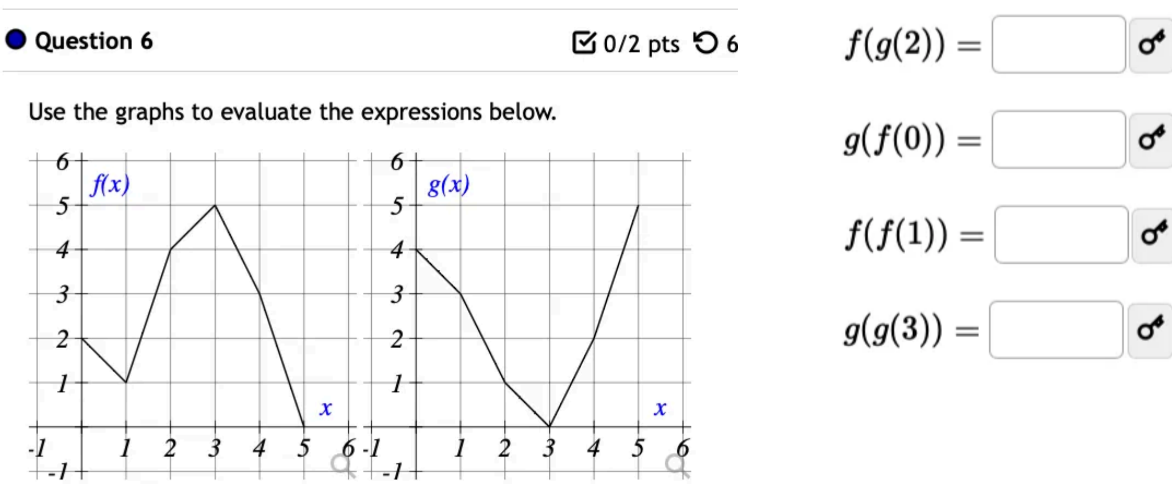
(classnotes) 3.4 Function Composition, part 2

$$(f \circ g)(x) = f(g(x))$$

Using graphs



Plugging in values

1. Given that $f(x) = 8x + 5$ and $g(x) = -x^2 - 2x + 16$, calculate:

- $f(g(2)) =$

- $g(f(7)) =$

> Self check: can you rewrite the $f(g(x))$ notation using $f \circ g(x)$ notation?

> Hint, for $f(g(2))$, break up the problem into 2 mini problems: first calculate $g(2)$, then use the result to plug into the $f(x)$ function!

2. Given that $f(x) = x^2 + 8x$ and $g(x) = x - 5$, calculate:

- $f(g(-3)) =$

- $\mathrm{f} \circ f(-3) =$

- $g \circ f(-3) =$

- $g \circ \mathrm{g}(-3) =$

Word Problem

The number of bacteria in a refrigerated food product is given by $N(T) = 23T^2 - 52T + 28$, where T is the temperature of the food.

When the food is removed from the refrigerator, the temperature is given by $T(t) = 8t + 1.3$, where t is the time in hours.

Find the composite function $N(T(t))$:

Find the time in hours when the bacterial count reaches 17483